



The effect of temperature on the geopolymerization process of a metakaolin-based geopolymer

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ABSTRACT

This paper describes the effect of different curing temperatures on the geopolymerization process, physical, mechanical and optical properties of a metakaolin-based geopolymer activated by alkali. The influence of different curing temperatures (within the range 30 to 90 °C) was studied systematically by means of differential scanning calorimetry (DSC), SEM, UV/Vis Spectrophotometry, Leaching analysis and Brunauer–Emmet–Teller method (BET). The results showed the existence of an optimum temperature at which the geopolymer presents the best physical and mechanical properties. The geopolymers cured at 30 and 90 °C presented high porosity, and were translucent to the Visible light, which makes possible to tailor this inorganic polymers for optical and photocatalytic applications.

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1. Introduction

Geopolymer is a broadly termed “inorganic polymer” [1,2]. In the synthesis of alkali-metakaolin system the geopolymerization stages can be supposed: (1) deconstruction (dissolution of metakaolin in alkaline solution), (2) polymerization of alumina/silica–hydroxy species and oligomers, and (3) stabilization, small gels formed are probably transformed into large networks through reorganization [3].

A supersaturated aluminosilicate solution (SAS) is created by dissolution [4] and the time for SAS to form a continuous gel varies considerably with raw materials, silica/alumina ratio and synthesis conditions [5–8]. Moreover, Geopolymers have received considerable attention [9–14], because of their low cost, excellent mechanical and physical properties, low energy consumption and reduced “greenhouse emissions” at the elaboration process. Therefore, the understanding of geopolymerization process can provide the key to control several factors such as the degree of polymerization, porosity along mechanical properties of geopolymers so that they may be tailored for specific applications. Besides, the most critical element of geopolymerization that has been briefly explored is the transformation from liquid precursor to

“solid” gel [13]. The majority of the literature about geopolymers deals with product characterization [4,6,7,9–11], and less effort has been made to understand the geopolymerization mechanism [3,5,14] leading to obtain a final specific characteristics of the geopolymer e.g. lightweight or dense and sound matrixes. Few reports [4,5,14] deal with the relationship between synthesis temperature and the kinetics of geopolymerization and its influence on physical and mechanical properties, along with lack of any qualitative or semi-quantitative description on the effect of unreacted raw material on those properties. In the present work the kinetics of geopolymerization of alkali–metakaolin system and the effect of a range of synthesis temperatures as well as the amount of total unreacted silicon on the physical and mechanical properties are investigated. Findings reported in this article would help to understand how the physical and mechanical properties of geopolymer materials can be tailored.

2. Materials and methods

Geopolymer samples were prepared by mechanically mixing distilled water, sodium hydroxide, sodium silicate ($\text{Na}_2\text{O}/\text{SiO}_2$ wt ratio: 0.31, SIDESA Corporation), and stoichiometric amounts of metakaolin (BASF Corporation) with molar ratio of $\text{Si}/\text{Al}=1$, surface area $13.0 \text{ cm}^2/\text{g}$ and mean particle size of $1.2 \mu\text{m}$. Samples were formulated with molar oxide ratios: $\text{SiO}_2/\text{Al}_2\text{O}_3=3.3$, $\text{Na}_2\text{O}/$

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$\text{SiO}_2 = 0.25$, $\text{Na}_2\text{O}/\text{Al}_2\text{O}_3 = 0.4883$, $\text{H}_2\text{O}/\text{Na}_2\text{O} = 13.73$. To remove entrained air, the samples were vibrated for 15 min and finally were transferred to cylindrical moulds covered with a film to prevent moisture loss. The geopolymer was cured in a laboratory oven in two steps: first, dried at 40 °C for 2 h, to prevent cracking due to an abrupt loss of water [1], second step, cured at 30, 40, 50, 60, 75 and 90 °C for 24 h, to develop mechanical properties.

The heat evolution of each geopolymer was determined for the isothermal experiments. Measurements were performed on a Mettler Toledo DSC822E (2 g of metakaolin were mixed with 3.2 ml of sodium silicate solution). For the calculation of the total heat evolution a straight baseline was used.

Porosity and pore size distribution were measured with a NOVA 2000e, Quantachrome-instruments, employing the BET method. SEM was performed to study the morphology of geopolymers cured at 30, 60 and 90 °C. Scatter transmittance of geopolymer cured at 30, 60 and 90 °C were measured with a Perkin Elmer UV/Vis/NIR spectrophotometer in combination with an integrating sphere. Samples were cut

to fit into the sample holder and wavelength was set to sweep the sample from 400 to 800 nm.

Ultimate compressive strength was determined according to standard ASTM C 733, using an universal testing machine (Tinius Olsen) with a capacity of 300 kN. Fifteen samples were tested. The mechanical strength of the samples was tested at the age of 7 day after casting. After this time, in the present work and elsewhere [7,8] the variation in compressive strength is not significant.

The amount of total unreacted silicon, NRS, (within metakaolin and sodium silicate) was determined on samples cured at 30, 60 and 90 °C. Each sample was ground and mixed with 12.5 g of geopolymer with 250 mL of KOH solution 2 M, to leach all the NRS. This procedure is similar to the one usually used to stop hydration in cements and was carried out at room temperature for 12 h, the time required to completely desegregate the sample. At the designated time intervals, 10 mL of the suspension were collected, centrifuged and filter to remove the remaining solids in the supernatant solution. The leached solution was used to analyze the Si compositions with a Perkin Elmer 3000 Inductively Coupled Plasma.

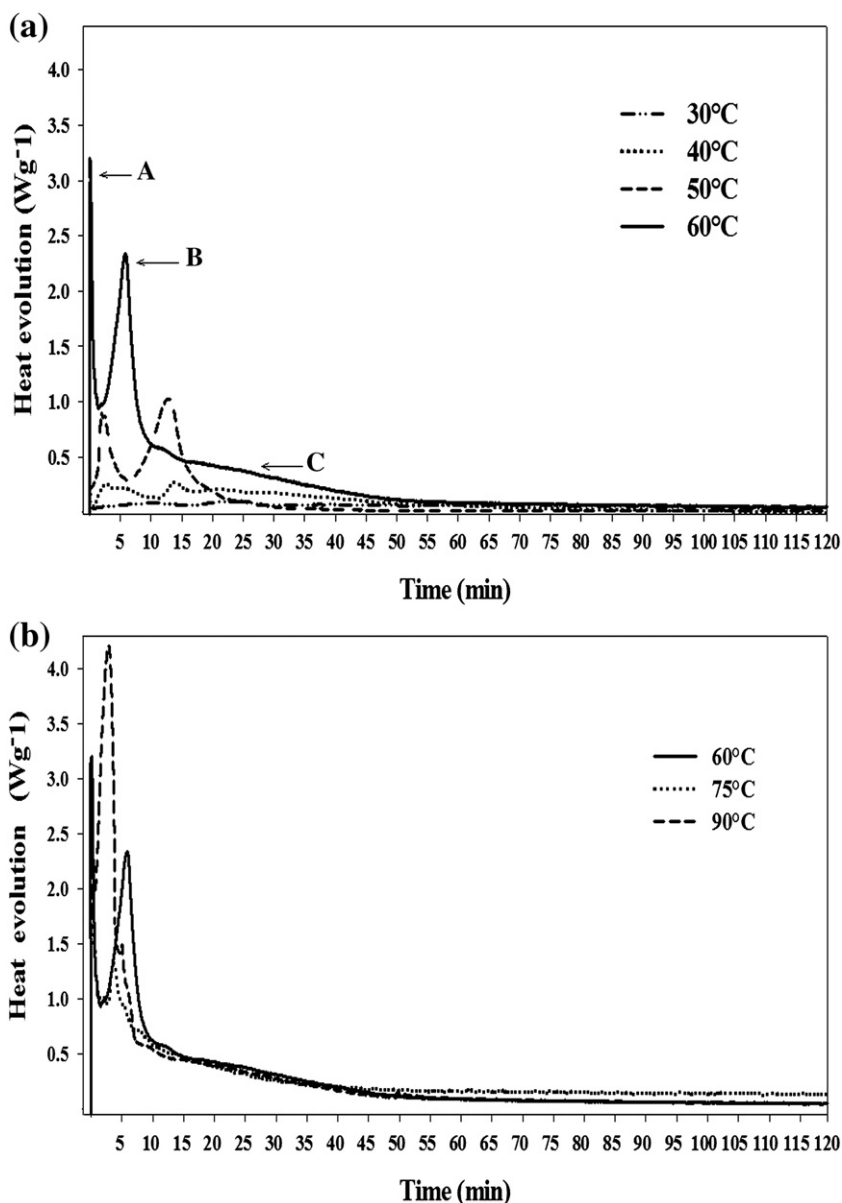


Fig. 1. Calorimetric results at different curing temperatures, (a) 30, 40, 50 and 60 °C (b) 60, 75 and 90 °C.

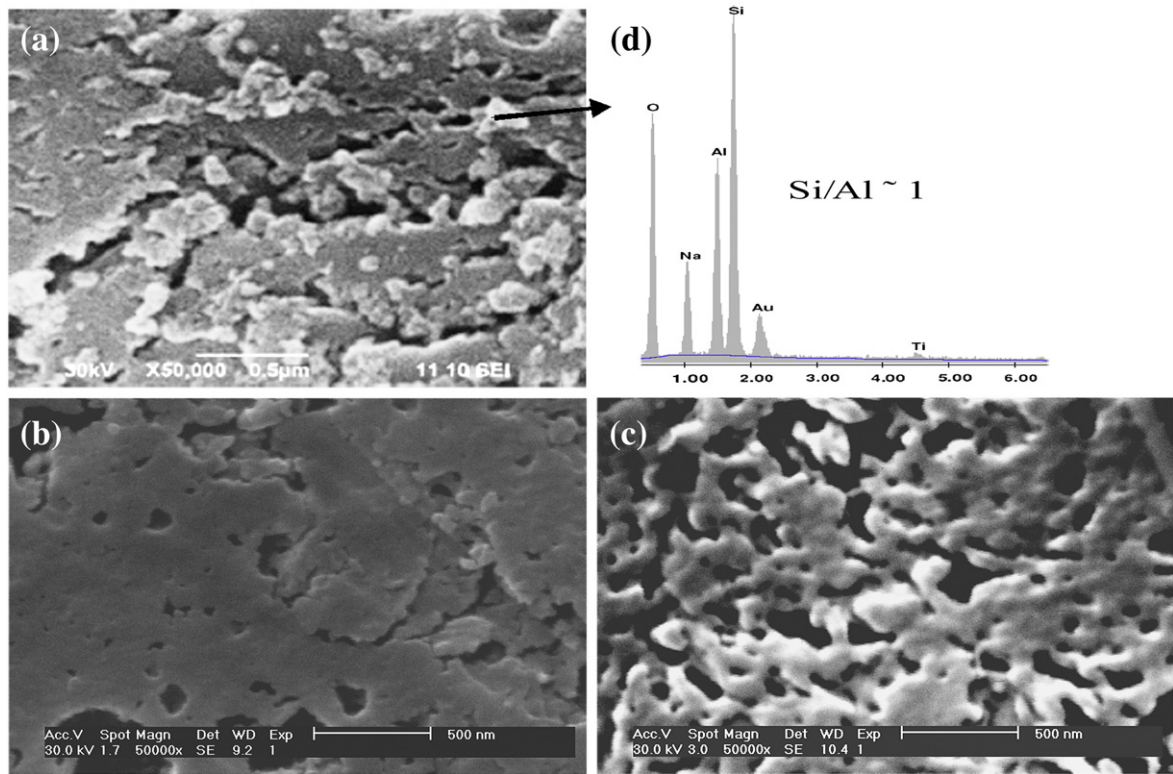


Fig. 2. SEM micrographs of the metakaolin-based geopolymer at curing temperatures of: a) 30 °C, b) 60 °C and c) 90 °C and (d) EDX analysis.

3. Results and discussion

3.1. Effect of curing temperature on geopolymerization process

DSC measurements were carried out at temperatures of 30, 40, 50, 60, 75 and 90 °C, Fig. 1(a). In general, the calorimetric curves show three peaks: peak A, the heat is evolved in a relatively short period of time, a dissolution of the solid metakaolin in the strong alkaline aqueous solution takes place; peak B a second exothermic reaction occurred suggesting the breaking down of metakaolin particles with the formation of oligomeric species in the aqueous phase in combination with the formation of few alumina/silica-hydroxy species [3,15]; peak C which is a small contribution of heat evolution (attributable to the polymerization/condensation reactions), implies that the process goes into a thermally steady stage and the large network consolidation [3,14]. Hence, it is qualitatively observed, how at 30, 40 and 50 °C at the first and second peak (Fig. 1(a)) the heat evolved is too short, meaning that the dissolution and the formation of hydroxy species and oligomers is not complete or inefficient while at 75 and 90 °C (Fig. 1(b)), the extent of the first and second peak is shortened suggesting that the time needed to break the metakaolin particles and to form the oligomer and alumina/silica-hydroxy species

is not enough. As a result of these, in both cases the polymerization or condensation reaction is not well established (since the reaction time appeared to be shortened), leading to low values of compressive strength. In addition, at 90 °C it is believed that the exothermic reactions would cause the losing of water more rapidly leading to the formation of micro-cavities which cause an increase in porosity, Fig. 2(c), hence lightweight products could be designed for specific applications.

On the other hand, the total heat evolution (after curing at 40 °C temperatures, the geopolymerization process can take place) decreases as the curing temperature increases, Table 1, implying

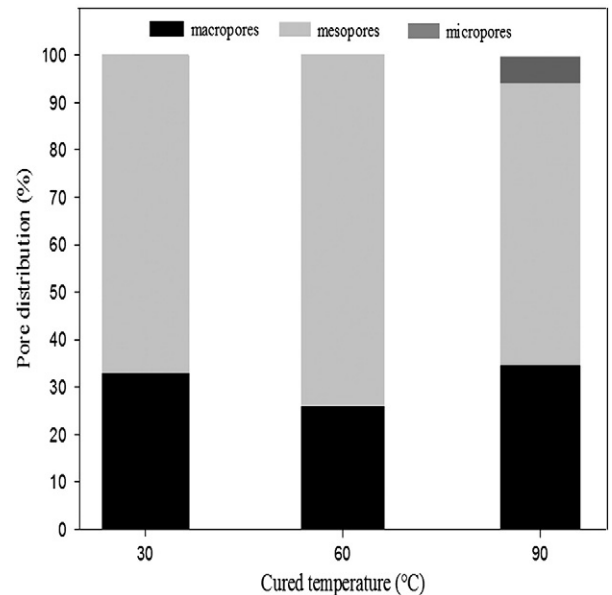


Fig. 3. Pore distribution of metakaolin-based geopolymer based cured at 30, 60 and 90 °C.

Table 1

The effect of curing temperature on compressive strength.

Reference		Property		
Step1	Step 2	Porosity (ccg ⁻¹)	Total heat evolution (Jg ⁻¹)	Compressive strength (MPa)
2 h, 40 °C	30 °C	0.2251 ± 0.001	497.07	7.032 ± 1.009
2 h, 40 °C	40 °C	0.2244 ± 0.005	1267.79	8.553 ± 1.326
2 h, 40 °C	50 °C	0.2038 ± 0.013	1414.58	11.771 ± 0.096
2 h, 40 °C	60 °C	0.1651 ± 0.020	1796.50	17.867 ± 1.009
2 h, 40 °C	75 °C	0.1710 ± 0.008	1480.82	13.771 ± 1.326
2 h, 40 °C	90 °C	0.1821 ± 0.002	1149.59	13.130 ± 0.496

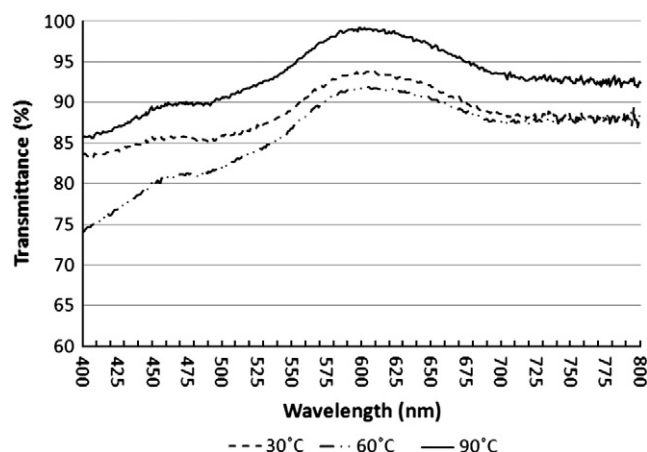


Fig. 4. Scatter Transmittance percent of geopolymers cured at 30, 60 and 90 °C.

that the geopolymerization extent is shorter. Nevertheless, at 60 °C the total heat evolved was the highest, 1796 J/g, which means that the stages or events in the geopolymerization process had occurred in a favorable manner. As a result of this, a more sound material is obtained with the highest compressive strength, Fig. 2(b) and Table 1.

3.2. Effect of curing temperature on porosity, total unreacted silicon and optical properties

Fig. 3 shows the pore size distribution according to IUPAC, Cioffi R. et al. [5]. The samples obtained at 60 °C have the minimum amount of macropores with the largest amount of mesopores. As a result of this, the compressive strength of the metakaolin-based geopolymer (Table 1) is strongly dependent on the size of porosity which in turn is the result of the geopolymerization process condition.

The mass percent of NRS at 30, 60 and 90 °C was: 51, 43 and 54, respectively. Hence, the high amount of NRS at 30 and 90 °C can be related to an inefficient dissolution and formation of hydroxy species and oligomers, therefore the polymerization or condensation reaction could not be established properly leading to low compressive strength and high porosity, Table 1. These may be ascribed to the fact that at 90 °C the reaction kinetics is high while the reaction time between surface layer of metakaolin and sodium silicate solution is long. It is believed that a large number of species Si^{4+} and Al^{3+} are formed and gelled quickly covering the surface of metakaolin, preventing their subsequent deconstruction. However, at 30 °C the reaction kinetics is very slow, Fig. 1(a), with a low absorption of the sodium silicate solution on the surface of the metakaolin (low dissolution of

metakaolin), besides, the entrapped water in pores prevents both gel and network formation, promoting the higher amount of macroporosity. In addition, unreacted metakaolin, Fig. 2(a) and (d), is a weak face which is expected to act as a point defect in the structure, intensifying locally the stress and precipitating failure [13]. Finally at 60 °C, the lowest porosity and NRS with highest compressive strength were obtained. However, geopolymers with high porosity, 30 and 90 °C, are translucent to the Visible light, Fig. 4, which open the possibility to tailor this inorganic polymers for optical and photocatalytic applications.

4. Conclusions

By using the isothermal calorimetric technique it was possible to identify the geopolymerization process of an alkali-metakaolin system at different curing temperatures. It was observed the existence of an optimum curing temperature, 60 °C to obtain the best geopolymerization process. This fact was reinforced by leaching analysis carried out on these inorganic geopolymers. As a result of this, the geopolymer obtained at this temperature exhibited the best physical and mechanical properties. The information gained contributes to a better understanding of the geopolymerization process and opens the possibility to design the synthesis of this kind of geopolymers for specific applications.

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